

Omega-3 Fatty Acids for the Treatment of Depression: Systematic Review and Meta-Analysis



We conducted a meta-analysis of randomized, placebo-controlled trials of omega-3 fatty acid treatment of major depressive disorder in order to determine efficacy and to examine sources of heterogeneity between trials. PubMed (1965-May 2010) was searched for randomized, placebo-controlled trials of omega-3 fatty acids for major depressive disorder. Our primary outcome measure was standardized mean difference in a clinical measure of depression severity. In stratified meta-analysis we examined the effects of trial duration, trial methodological quality, baseline depression severity, diagnostic indication, dose of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) in omega-3 preparations, and whether omega-3 fatty acid was given as monotherapy or augmentation. In 13 randomized, placebo-controlled trials examining the efficacy of omega-3 fatty acids involving 731 participants, meta-analysis demonstrated no significant benefit of omega-3 fatty acid treatment compared to placebo (SMD=0.11, 95% CI: -0.04, 0.26). Meta-analysis demonstrated significant heterogeneity and publication bias. Nearly all evidence of omega-3 benefit was removed after adjusting for publication bias using the trim-and-fill method (SMD=0.01, 95% CI: -0.13, 0.15). Secondary analyses suggested a trend towards increased efficacy of omega-3 fatty acids in trials of lower methodological quality, trials of shorter duration, trials, which utilized completers rather than intention-to-treat analysis, and trials in which study participants had greater baseline depression severity. Current published trials suggest a small, non-significant benefit of omega-3 fatty acids for major depression. Nearly all of the treatment efficacy observed in the published literature may be attributable to publication bias.